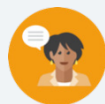


6.7 Triangle Similarity

Lesson Introduction

 Benchmark	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.		 Learning Targets	- I can identify similar triangles and write similarity statements. - I can use given information to show that two triangles are similar.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- What makes two triangles similar? - How can you determine if two triangles are similar if you do not know any angle measures? - How can you determine if two triangles are similar if you do not know any side lengths?			
 Lesson Narrative	This lesson offers students the opportunity to explore different ways of proving two triangles similar. Students will explore this through a sequence of transformations as well as by creating triangles following a given description. The formalized names for the theorems that demonstrate similarity are identified and connections are made to the drawings.			
 Component Summary	6.7.1 Warm-Up (~5 min.)	Describing a sequence of transformations (<i>SE p. 326</i>)	6.7.3 Your Turn! (5 min.)	Determining similarity (<i>SE p. 328</i>)
	6.7.2 Exploration (10 min.)	Describing a sequence of transformations that show similarity (<i>SE p. 327</i>)	6.7.4 Activity (20-25 min.)	Creating triangles given a description to determine similarity (<i>SE p. 329</i>)
	6.7.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 331</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Angle-Angle Similarity (AA~) Theorem - Side-Side-Side (SSS~) Theorem - Side-Angle-Side (SAS~) Theorem <u>Additional Terms:</u> - Similarity - Congruent - Proportional		- Centimeter Rulers - Protractor (Optional) Patty Paper (Optional) Chart Paper	6.7.4 Activity includes a Gallery Walk that requires chart paper for each station. Print the instructions and post at each station. - Consider verifying the measurements for 6.7.2 Exploration.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not state that 6.7.4 Activity descriptions D and F do not create triangles. - Students may not understand that if the scale factor is greater than 1, then the dilated image is larger than the original figure and if the scale factor is between 0 and 1, then the dilated image is smaller than the original figure.	- For the 6.7.4 Activity, consider adding a step of writing similarity statements for each pair of triangles that are similar. Students will first need to label the vertices of the triangles prior to doing this. - Consider only giving descriptions A-E to all students for 6.7.4 Activity and using F and G for those groups who finish early.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Group Presentations 6.7.4 Activity provides an opportunity for students to work in partner pairs or small groups to draw triangle pairs based on given descriptions. For question 3, student groups can present one of the drawings to the whole group and provide justification of why they are or are not similar and what theorem matches the description. Ask class members to restate their understanding of the group’s reasoning.	MA.K12.MTR.4.1: Discussion and Reflection In each component of this lesson, students discuss and reflect on mathematical topics and ideas. 6.7.2 Exploration provides students an opportunity to explore similar triangles. 6.7.4 Activity also encourages students to work through generating similar figures using given information. Teachers should consider highlighting common misconceptions and revelations students had to further deepen student understanding.
	6.7.4 Activity is a great opportunity for providing multiple entry-points to the mathematical language and concepts. In question 1, students first draw the pairs of triangles for a kinesthetic entry-point. In question 2, students can visually see the possibilities of similarity by observing their drawings and those of their partner and/or other partner pairs. Then, in question 3 students connect the provided theorems to what they have observed in the triangle relationships.	
Lesson Notes:		

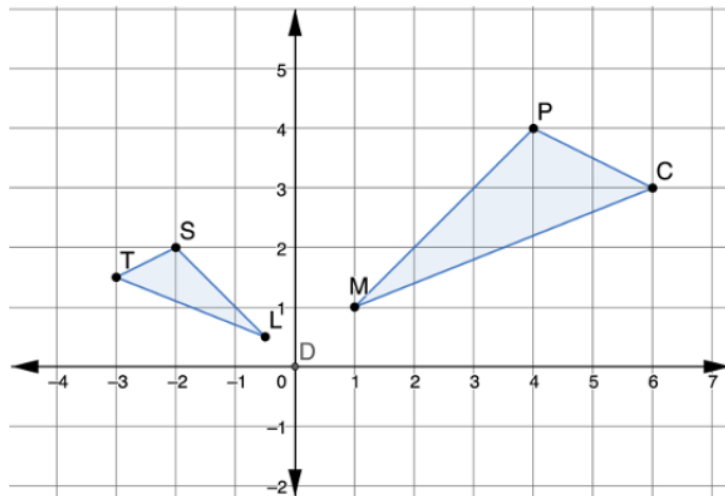
6.7.1 Warm-Up: Bell Work

Component Overview: Students describe a sequence of transformations that will map one triangle onto the other.



6.7.1 Warm-Up: Bell Work

1. $\triangle MPC$ and $\triangle TSL$ are shown on the coordinate grid, where $\triangle MPC$ is the pre-image.



Describe a sequence of two transformations that will map the triangles.

Answers will vary. Dilate $\triangle MPC$ using a scale factor of $\frac{1}{2}$ centered at point D. Then, reflect $\triangle M'P'C'$ across the y-axis.



Are the transformations in your sequence commutative for mapping this preimage onto the image?



Consider providing students with patty paper to help them determine the rigid motion(s) in the sequence.

6.7.2 Exploration: Similar Triangles

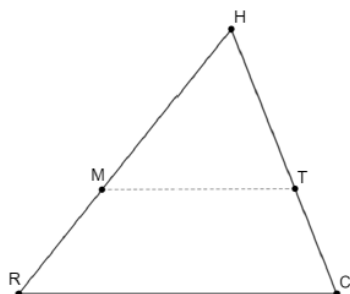
Component Overview: Students will explore the connection between the ratio of side lengths in similar triangles. Once it is determined that the provided triangles are similar, they list all transformations that will result in similar triangles.



6.7.2 Exploration: Similar Triangles

With your partner, complete the following.

$\triangle HCR$ is shown, where $m\angle HCR = 68^\circ$, $m\angle HRC = 51^\circ$, and $m\angle RHC = 61^\circ$. Point M is on $\overline{HR}$ and point T is on $\overline{HC}$. $\overline{MT}$ is drawn as a dotted line.



- Use a ruler and a protractor to complete the table.
Answers will vary for length measurements depending on the version used in print.

Measurements of $\triangle RHC$		Measurements of $\triangle MHT$	
$RH = 7.7 \text{ cm}$	$m\angle HRC = 51^\circ$	$MH = 4.2 \text{ cm}$	$m\angle HMT = 51^\circ$
$HC = 6.5 \text{ cm}$	$m\angle RHC = 61^\circ$	$HT = 3.9 \text{ cm}$	$m\angle MHT = 61^\circ$
$RC = 7.2 \text{ cm}$	$m\angle HCR = 68^\circ$	$MT = 4.3 \text{ cm}$	$m\angle HTM = 68^\circ$

- What do you notice about the corresponding angles of $\triangle RHC$ and $\triangle MHT$?
These angles are the same measure.



Throughout 6.7.2 Exploration, listen carefully for key words like “ratio,” “proportion,” “similar,” “congruent,” and “angles” to ensure that mathematical discussion is taking place and that students are reflecting on the questions.



Consider giving students a larger triangle on a separate sheet of paper created using GeoGebra or Desmos. Use the provided angle measurements to create the triangle.

6.7.2 Exploration: Similar Triangles (continued)

3. Discuss with your partner what relationship could have been given so that it would not have been necessary to use a protractor to find $m\angle HMT$ and $m\angle HTM$. Summarize your discussion.

Answers will vary. If we had been given that $\overline{RC} \parallel \overline{MT}$ then the angles would have been congruent by corresponding angles.

4. What do you notice about the corresponding sides of $\triangle RHC$ and $\triangle MHT$?

Answers will vary. The corresponding sides are proportional.

5. Discuss with your partner how $\triangle RHC$ and $\triangle MHT$ are related. Summarize your findings.

Answers will vary. All the corresponding angles of the triangles are congruent and all of the corresponding sides are proportional.

6. Complete the statement that shows the relationship. $\triangle RHC$ is similar to $\triangle MHT$.

7. Complete each relationship for $\triangle RHC$ and $\triangle MHT$.

$$\frac{RH}{MH} = \frac{RC}{TM}$$

$$\frac{HT}{HC} = \frac{TM}{CR}$$

8. Discuss with your partner a sequence of transformations that can be used to map $\triangle RHC$ onto $\triangle MHT$. Summarize your discussion by describing the sequence.

Answers will vary. Dilate $\triangle RHC$ using a scale factor of $\frac{3}{5}$ centered at point H .



Question 3 offers a moment for spiraling back to Unit 3. While monitoring students, look and listen for use of the vocabulary like “parallel lines” and “corresponding angles.” For students not using the vocabulary, ask probing questions to guide them back to this vocabulary.



When describing the relationship between the two triangles, students may write the reciprocal of the scale factor needed depending on the wording of their answer. Question students that do this by asking what happens to values when multiplying by fractions greater than 1.

6.7.2 Exploration: Similar Triangles (continued)

9. Compare your sequence of transformations with another pair of students. Ask two classmates for their sequence of transformations. Record their sequence and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating your summary is correct.

Answers will vary.

Sequence	My Summary of Their Reason	Initials

10. What transformation ensures that the sequence of transformations will result in similar triangles?
The dilation ensures that the result is similar triangles.



Observe and listen as students work with their partners. Make note of student ideas that you can ask to be shared in the whole group before beginning 6.7.3 Your Turn! Consider highlighting student voices to share rich discussions or comments overheard while you circulated or to address common misconceptions that were evident among the majority of the class.

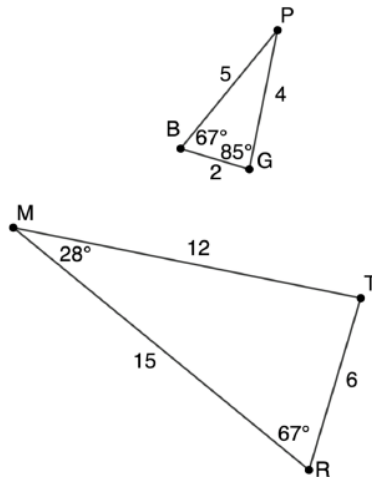
6.7.3 Your Turn!

Component Overview: Students determine if two triangles are similar. If they are, students write a similarity statement. If they are not, students provide an explanation of why not.



6.7.3 Your Turn!

1. $\triangle PGB$ and $\triangle MTR$ are shown. Use the given information to determine if $\triangle PGB$ is similar to $\triangle MTR$. If they are similar, write a similarity statement. If they are not, explain why.



Yes. $\triangle PGB \sim \triangle MTR$.



While monitoring students, make note of different methods of determining similarity. When discussing this question as a group, call on students to share their different strategies of thinking. Hearing different methods from peers can help resolve misconceptions that some students have.

6.7.4 Activity: Stations

Component Overview: Students will draw pairs of triangles from given descriptions. Provide an opportunity for students to observe the drawings of other partner groups to make connections to which descriptions produced similar triangles. Students will then name the similarity theorems for all descriptions that produced similar images.



6.7.4 Activity: Stations

1. With your partner, create two triangles using the description provided. Try to make the two triangles as different as possible.

Description Letter	Drawing
A	
B	



When drawing the triangles, if the side measures are not given in the description, students should draw their triangles with sides no shorter than 2 cm.

After the triangles are drawn using the descriptions, students should then label all of the sides and angles with their measurements.

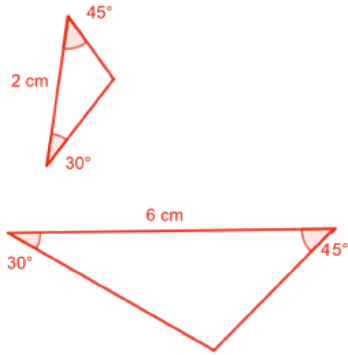
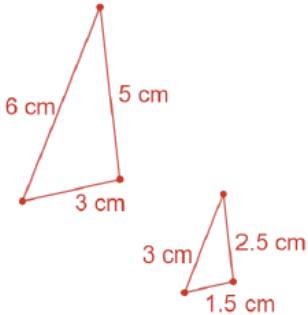


6.7.4 Activity can be done on large poster or chart paper, as stations that partner groups move through, or with activity cards that are passed from partner group to partner group. Consider printing multiple sets of descriptions and dividing the class into multiple partner groups.

Consider only giving descriptions A-E to all students for 6.7.4 Activity and using F and G for those groups who finish early.

Students will need a ruler with centimeters and a protractor.

6.7.4 Activity: Stations (continued)

Description Letter	Drawing
C	
D	

6.7.4 Activity: Stations (continued)

Description Letter	Drawing
E	
F	

6.7.4 Activity: Stations (continued)

Description Letter	Drawing
G	The drawing shows two triangles. The top triangle has angles of 105°, 45°, and 30°. The bottom triangle has angles of 105°, 45°, and 30°.



Creating triangles with only angles can be difficult. If students struggle, consider suggesting they draw the three angles separately on patty paper, cut them out, and then slide the pieces together to create the triangles.

2. Which descriptions resulted in similar triangles?
B, C, D, F, and G are similar figures.

6.7.4 Activity: Stations (continued)



Angle-Angle (AA~) Theorem

If two angles of one triangle are congruent to two angles of another triangle, then the two triangles are similar.



Side-Side-Side Similarity (SSS~) Theorem

If the lengths of the corresponding sides of two triangles are proportional, then the triangles are similar.



Side-Angle-Side Similarity (SAS~) Theorem

If the lengths of two sides are proportional and their included angles are congruent on two different triangles, then the triangles are similar.



Consider having students develop a Foldable, Frayer Model, or Word Wall that includes all these terms. Possibly categorize these theorems with others they have learned throughout the course so far.



What similarities and differences do you notice between these similarity theorems and the congruence theorems learned in Unit 5?

3. Match each description with the postulate or theorem that describes it. If none describe it, then write "none."

Description Letter	Theorem
A	None
B	Angle-Angle Similarity
C	Angle-Angle Similarity
D	Side-Side-Side Similarity
E	None
F	Side-Angle-Side Similarity
G	Angle-Angle Similarity



To provide a challenge, for the descriptions that did not result in similar triangles, have students write what additional information that, if provided, would have resulted in similar triangles.

6.7.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



6.7.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.
Answers will vary.



This would be a great opportunity to explore more about what the students may still have questions about or predict where this topic may go next. Have students predict questions they may see in the next lesson or on the next assessment.